CSC5010 Project 3 Report: Transformer Text Classification

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1. Project Goal

This project runs a Transformer encoder classifier on a truncated Chinese THUCNews dataset. The task is to classify Chinese news titles into 10 categories: 财经, 房产, 股票, 教育, 科技, 社会, 时政, 体育, 游戏, and 娱乐.

The grading requirements define two experiments. Task A uses the original text without preprocessing. Task B applies text preprocessing before training and testing. Both tasks must output running results and misclassified documents, then compare cases where one task is correct and the other is wrong.

2. Dataset and Model

- Training set: 45,000 titles
- Development set: 2,500 titles
- Test set: 2,500 titles
- Vocabulary limit: 10,000 characters plus `<UNK>` and `<PAD>`
- Sequence length: 32 characters
- Model: Transformer encoder classifier
- Epochs: 6
- Seed: derived from student ID 225040065

The model uses a character-level tokenizer. Each title is padded or truncated to 32 characters. The Transformer includes token embedding, sinusoidal positional encoding, multi-head self-attention, feed-forward layers, residual connections, layer normalization, and a final linear classifier.

3. Task A: Original Text

Task A uses the original dataset directly. No punctuation or whitespace is removed before building the vocabulary or encoding the train/dev/test sets.

- Test loss: 0.493174
- Test accuracy: 0.846800
- Misclassified test documents: 383

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Test Accuracy: 0.846800
Test Loss: 0.493174
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The full Task A classification report and confusion matrix are saved in `outputs/taskA/taskA_metrics.txt`. All predictions are saved in `outputs/taskA/taskA_predictions.csv`, and all wrong predictions are saved in `outputs/taskA/taskA.misclassified.csv`.

4. Task B: Preprocessed Text

Task B removes Chinese and English punctuation and normalizes whitespace before vocabulary construction and encoding. The labels and train/dev/test split are unchanged. This keeps the experiment focused on the effect of text cleaning.

- Test loss: 0.570312
- Test accuracy: 0.842400
- Misclassified test documents: 394

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Test Accuracy: 0.842400
Test Loss: 0.570312
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The full Task B classification report and confusion matrix are saved in `outputs/taskB/taskB_metrics.txt`. All predictions are saved in `outputs/taskB/taskB_predictions.csv`, and all wrong predictions are saved in `outputs/taskB/taskB.misclassified.csv`.

5. Task A vs. Task B Comparison

Task	Preprocessing	Test accuracy	Misclassified documents
Task A	None	0.846800	383
Task B	Remove punctuation / normalize whitespace	0.842400	394

Task A wrong but Task B correct: 138 documents.
Task B wrong but Task A correct: 149 documents.

6. Three Documents Wrong in Task A but Correct in Task B

Test index	Document	True label	Task A prediction	Task B prediction
5	本科未录取还有这些路可以走	教育	房产	教育
28	调查显示：29.5%的人不满意当年所选高考专业	教育	股票	教育
105	五W让你提高英语四六级听力	教育	股票	教育

Brief explanations:

1. `本科未录取还有这些路可以走`: The cleaned version keeps the main words but changes token positions after normalization, which can alter attention patterns.
2. `调查显示：29.5%的人不满意当年所选高考专业`: Task B removes 3 punctuation/space characters, making the core keywords more concentrated for the character-level Transformer.
3. `五W让你提高英语四六级听力`: The cleaned version keeps the main words but changes token positions after normalization, which can alter attention patterns.

7. Three Documents Wrong in Task B but Correct in Task A

Test index	Document	True label	Task A prediction	Task B prediction
43	教育话题：不能让何川洋做“牺牲品”	教育	教育	财经
67	名校农村生比例降低 该酝酿“平权法案”了	教育	教育	社会
89	浙商大法语系介绍：半数毕业生出国深造	教育	教育	社会

Brief explanations:

1. `教育话题：不能让何川洋做“牺牲品”`: The removed punctuation may have carried useful title structure; after cleaning, 3 characters are removed and the model loses some boundary cues.
2. `名校农村生比例降低 该酝酿“平权法案”了`: The removed punctuation may have carried useful title structure; after cleaning, 3 characters are removed and the model loses some boundary cues.
3. `浙商大法语系介绍：半数毕业生出国深造`: The removed punctuation may have carried useful title structure; after cleaning, 1 characters are removed and the model loses some boundary cues.

8. Output Files

- `outputs/taskA/taskA_metrics.txt`: Task A running logs, loss, accuracy, classification report, and confusion matrix.
- `outputs/taskB/taskB_metrics.txt`: Task B running logs, loss, accuracy, classification report, and confusion matrix.
- `outputs/taskA/taskA_predictions.csv`: all Task A test predictions.
- `outputs/taskB/taskB_predictions.csv`: all Task B test predictions.
- `outputs/taskA/taskA.misclassified.csv`: all Task A misclassified test documents.
- `outputs/taskB/taskB.misclassified.csv`: all Task B misclassified test documents.
- `outputs/taskA_wrong_taskB_correct.csv`: documents fixed by preprocessing.
- `outputs/taskB_wrong_taskA_correct.csv`: documents hurt by preprocessing.

9. Conclusion

The project successfully runs the Transformer classifier for both required tasks and outputs the requested misclassified documents. Task B tests whether punctuation removal and whitespace normalization improve the model. Comparing the cross-corrected documents shows that preprocessing can help when punctuation distracts from core category words, but it can also hurt when punctuation provides useful boundary or title-structure cues. The final deliverable includes this report, all source code, all output results, and the generated comparison files.